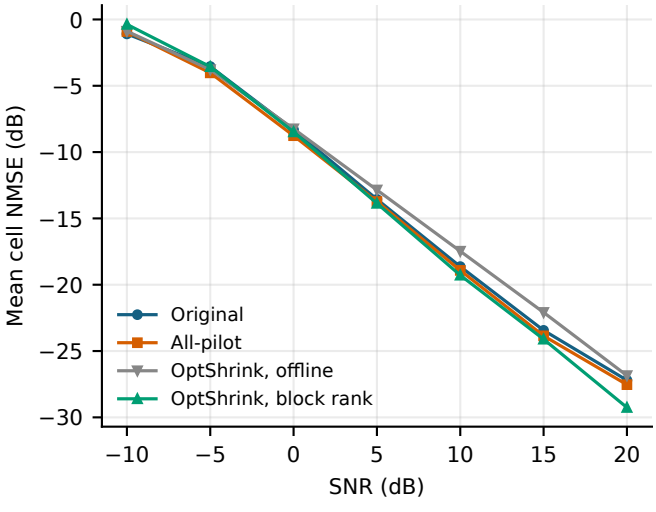
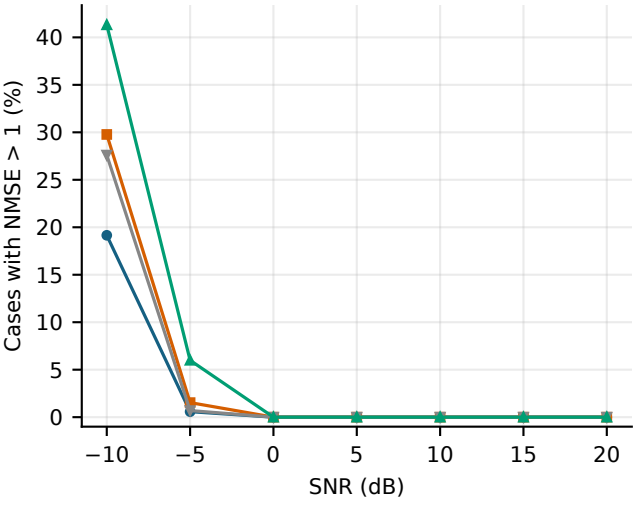

Mean cell NMSE (dB)
0
−5
−10
−15
−20
−25
−30
−10
−5
0
5
10
15
20
SNR (dB)
Original
All-pilot
OptShrink, offline
OptShrink, block rank


Cases with NMSE > 1 (%)
40
35
30
25
20
15
10
5
0
−10
−5
0
5
10
15
20
SNR (dB)